MMAGL: Multiobjective Multiview Attributed Graph Learning for Joint Clustering of Hyperspectral and LiDAR Data

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***Abstract*— The joint clustering of multimodal remote sensing (RS) data represents a multiobjective optimization challenge involving conflicting modality-specific objectives and diverse regularization objectives. Current approaches to multiview subspace clustering (MVSC) often oversimplify this task by transforming it into a weighted single-objective optimization problem, neglecting the intricate interactions between objectives and leading to suboptimal subspace representations. The presence of quadratic decision variables in MVSC renders direct application on large-scale RS data impracticable using multiobjective evolutionary algorithms (MOEAs). To overcome this challenge, we propose a novel MVSC method termed multiobjective multiview attributed graph learning (MMAGL). Instead of optimizing every self-representation coefficient individually, our method transforms MVSC into a link prediction task over a sparse attributed graph that fuses different modalities. We incorporate superpixel-based sample reduction and proximity-based population coding, leveraging spatial and structural priors, respectively. This results in a significantly compressed decision space, enabling optimization with MOEAs. To fully exploit node attributes and the graph structure, we redefine self-representation using contrastive learning and introduce an efficient graph filtering (GF) through a generalized spectral graph convolution, enhancing clustering discriminability. The proposed MMAGL constitutes a hybrid and versatile framework, adaptable to any MOEA. Extensive experimental evaluations demonstrate that our MMAGL method surpasses the current state-of-the-art on multimodal RS benchmarks (e.g., with nearly 2% gain on Trento and 3% on Houston) on overall accuracy.**

***Index Terms*— Graph learning, hyperspectral and light detection and ranging (LiDAR) data, multiobjective optimization, multiview subspace clustering (MVSC).**

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NOMENCLATURE

N	Number of pixels.
V	Number of modalities.
v	vth view/modality.
M	Number of anchors.
K	Number of clusters.
D_v	Feature dimensionality of the vth modality.
$\mathbf{X}^v$	Data matrix for the vth modality.
$\mathbf{Z}$, $\mathbf{Z}^v$, $\mathbf{Z}^*$	Subspace representation coefficient matrices.
$F(\cdot)$	Multiobjective optimization problem.
$f_i(\cdot)$	ith objective function.
$\mathcal{S}$	Set of homogeneous superpixels.
M	Number of superpixels.
$\mathbf{Q}$	Association matrix of superpixels and pixels.
$\mathbf{X}_H$, $\mathbf{X}_L$	Node feature matrix of hyperspectral and LiDAR.
$\mathcal{N}_k(\boldsymbol{x}_i)$	Neighbor set of $\boldsymbol{x}_i$.
$\mathcal{G}_H$, $\mathcal{G}_L$	Attribute graph of hyperspectral and LiDAR.
$\mathbf{A}_H$, $\mathbf{A}_L$	Adjacency matrix of hyperspectral and LiDAR.
$\mathbf{H}_H$, $\mathbf{H}_L$	Graph embedding matrix of hyperspectral and LiDAR.
$\mathbf{L}_H$, $\mathbf{L}_L$	Graph Laplacian matrix of $\mathbf{A}_H$ and $\mathbf{A}_L$.
$\mathcal{G}_F$	Fused attribute graph.
$\sigma_i(\mathbf{L})$	ith eigenvalue of $\mathbf{L}$.
$\tilde{\mathbf{X}}$	Reconstructed feature matrix.
$\hat{\mathbf{X}}$	Augmented feature matrix.
$\boldsymbol{u}$	Decision vector of MOEA.
α	Coefficient in graph embedding.
λ	Eigenvalue of graph Laplacian matrix.

I. INTRODUCTION

LAND-COVER classification, the process of assigning specific labels to each pixel in remotely sensed images, stands as a fundamental challenge in the

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Earth observation (EO) community. With the evolution of EO techniques, multimodality-based land-cover classification has emerged as a promising avenue of research. Specifically, the fusion of hyperspectral (HS) data [1], [2], offering rich content information about surface materials, and light detection and ranging (LiDAR) data [3], providing elevation information, facilitates the discrimination of hard objects with subtle spectral differences [4].

Over recent decades, supervised paradigms have prevailed in multimodal joint classification of HS and LiDAR data [5], [6], achieving notable success with both shallow and deep methods as reviewed in [7]. However, these methods rely heavily on sufficient training labels, particularly in deep learning approaches, resulting in labor-intensive and costly efforts [8]. As an alternative, there is a growing interest in unsupervised classification of multimodal remote sensing (RS) data, known as multiview clustering (MVC) in the machine learning community [9], [10], differing from traditional single-source clustering [11]. Yet, the inherent heterogeneity of multimodal RS data poses significant challenges in measuring sample similarity and fusing complementary information, crucial elements in MVC.

Multiview subspace clustering (MVSC) [12], [13] has emerged as an effective paradigm for multimodal clustering due to its ability to capture intrinsic semantic structures. MVSC assumes that all multimodal inputs share a consensus low-dimensional subspace representation. This assumption proves beneficial for modeling interdependence between different land covers [14]. Methodologically, existing MVSC methods formulate the problem into a weighted single-objective optimization problem [15], [16]

$$\min \sum_{v} (\mu_v \cdot \Delta_v + \beta_v \cdot \Omega_v) + \lambda \cdot \Psi \tag{1}$$

where Δ_v, Ω_v, and Ψ represent view-specific self-representation error, view-specific regularization, and subspace fusion terms, respectively. These terms are weighted by tuneable hyperparameters $\{\mu_v, \beta_v\}$ and λ. While MVSC optimization typically relies on an alternating optimization strategy, and the semantic labels are obtained through spectral clustering over the consensus subspace representation, traditional MVSC methods suffer from three key drawbacks that hinder their application to multimodal RS data.

First, existing MVSC methods often amalgamate conflicting sub-objectives (e.g., Δ_v, Ω_v, and Ψ) into a single-objective optimization problem [17], necessitating the introduction of numerous hyperparameters (e.g., μ_v, β_v, and λ). This approach requires expert knowledge for hyperparameter adjustment and achieves only locally optimal solutions. A more straightforward solution to address this issue is to model MVSC as a multiobjective optimization problem (MOP). On this basis, one can use multiobjective optimizers, such as MOEAs [18], to pursue a global Pareto optimal solution set. However, quadratic decision variables involved in MVSC prevent the practical application of MOEAs. As a result, limited efforts have been made on multiobjective MVSC due to its expansive search space, especially compared to unimodal subspace clustering [19], [20].

Second, traditional MVSC methods exhibit high computational and space complexity [21], making them impractical for large-scale RS datasets. To address this challenge, a common approach is to adopt anchor graphs [16], [22] in lieu of full graphs, enabling scalable spectral segmentation. Although this strategy significantly reduces computational complexity [23], its effectiveness relies on a heuristic process for anchor point selection. Moreover, these compromise methods often overlook spatial properties in RS data, such as homogeneity. In the multimodal fusion of RS data, spatial priors play a crucial role in modeling complex data distributions.

Third, traditional MVSC solely focuses on the reconstruction similarity of each data point with itself, overlooking broader relationships with others and resulting in high sensitivity to outliers. Robust clustering models, as proposed in [24], necessitate substantial interdependence, often realized through a predefined graph. Consequently, there is a growing interest in integrating subspace clustering with attributed graphs [25], [26], incorporating spectral graph convolutions [27]. Recent advancements in metric learning [28], [29] highlight the significant improvement in discriminative power achieved through both positive pairwise relationships and negative ones, aligning well with clustering tasks. For example, Cai et al. [25] developed neighborhood contrastive subspace clustering based on this observation.

To address these limitations, this article introduces multiobjective multiview attributed graph learning (MMAGL), a general framework for the joint clustering of hyperspectral and LiDAR data. The key components of MMAGL are as follows. First, MVSC is reformulated as a multiview attributed graph clustering task by constructing view graphs. This enables the utilization of structural priors, where we define low-pass graph filtering (GF) on the attributed graph to aggregate nonlocal neighboring information. Second, we formulate MVSC as an MOP, encompassing view-specific reconstructions for hyperspectral and LiDAR data, as well as sparsity and block diagonalization of subspace representation. Given the widespread availability of MOEA, it is an ideal choice for solving the problem. However, the consensus subspace representation poses a challenge due to its undesirable high dimensionality for MOEA methods, resulting in an expanded search space with slow convergence. To mitigate this, we introduce two efficient strategies, i.e., superpixel-level graphs and proximity-based population coding, to reduce the search space. This ensures the feasibility of seeking the optimal consensus subspace representation through an MOEA method. Lastly, negative pair-wise relationships are incorporated into view-specific self-representation using contrastive learning, thereby enhancing the discriminative power and robustness of the subspace representation.

The main contributions of this article are succinctly summarized as follows.

1) We present MMAGL, a novel multiview attributed graph clustering framework, utilizing multiobjective optimization. MMAGL learns a globally optimal consensus subspace from hyperspectral and LiDAR data, incorporating spatial and structural priors simultaneously.

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2) A generalized GF operation is presented, featuring a closed-form solution to generate discriminative graph embeddings. Furthermore, we define a neighborhood contrastive self-representation (NCSR) on these embeddings, contributing to the improvement of clustering robustness.
3) Leveraging the distinctive characteristics of both RS images and subspace structure, we propose several strategies to reduce the search space for the proposed MOP. These strategies enhance the efficiency of the MOEA methods.
4) Extensive experiments conducted on diverse multimodal RS data validate that MMAGL surpasses state-of-the-art methods. Consequently, MMAGL offers an intriguing alternative for addressing multimodal RS clustering problems.

The remainder of this article is organized as follows. Section II provides a brief review of related works on MVC and multiobjective optimization. In Section III, we describe the details of the proposed MMAGL framework. Section IV is dedicated to presenting the results of systematic experiments and their corresponding analyses. Finally, our findings are summarized and discussed in Section V, offering insight derived from the study.

II. Preliminaries

In this article, we employ lower-case bold letters for vectors and upper-case bold letters for matrices. Specifically, $\mathbf{I}$ denotes an identity matrix. The transpose and trace of matrix $\mathbf{A}$ are denoted as $\mathbf{A}^\top$ and $\mathrm{Tr}(\mathbf{A})$, respectively. Nomenclature summarizes main notations utilized throughout this article.

A. Multiview Clustering

MVC aims to categorize data points into distinct groups based on their heterogeneous features. The effectiveness of designing MVC methods hinges on the fusion of complementary information and the measurement of similarity for multimodal inputs. In recent decades, subspace-based fusion has garnered increased interest, inspiring numerous notable works [10], commonly known as MVSC [12]. Other promising alternatives include multi-kernel learning [30], graph-based fusion [23], and deep learning-based fusion [31].

Let $\{\mathbf{X}^v \in \mathbb{R}^{D_v \times N}\}_{v=1}^V$ represent the multiview inputs, where N, D_v, and V denote the number of samples, dimensions, and views, respectively. Formally, MVSC typically seeks consensus subspaces, which can be represented as follows:

$$\arg\min_{\mathbf{Z}^v, \mathbf{Z}^*} \sum_{v=1}^{V} \mu_v \|\mathbf{X}^v - \mathbf{X}^v\mathbf{Z}^v\|_F^2 + \beta_v \|\mathbf{Z}^v\|_1 + \lambda \sum_{v=1}^{V} \left\|\mathbf{Z}^* - \mathbf{Z}^v\right\|_F, \quad \text{s.t. } \mathbf{Z}^v,\ \mathbf{Z}^* \in \mathcal{C}. \tag{2}$$

Here, $\mathbf{Z}^v$ represents the view-wise subspace representation matrix, $\mathbf{Z}^*$ is the consensus subspace representation across multiple modalities, and $\mathcal{C}$ denotes the constraints.

Despite the remarkable success of MVC methods, they have not received commensurate attention in the field of multimodal RS. Only a few works [16], [32] have attempted to apply MVC to multimodal RS data, and their clustering performance lacks competitiveness compared with the current state-of-the-art results. There are several shortcomings when applying MVSC to the multimodal RS domain.

1) MVSC overlooks the interaction of different learning objectives when combining them into a single-objective problem, posing challenges in adjusting the corresponding hyperparameters.
2) Existing MVSC cannot handle large-scale multimodal RS data due to its high computational complexity.
3) The classic self-representation method faces insufficiency in exploring negative relationships between samples.

To address these challenges, this article introduces a novel hybrid framework that reconceptualizes MVSC as an MOP, as opposed to merging MVSC into a weighted single-objective problem. This framework enables us to unveil the intricate interactions among sub-objectives in MVSC, facilitating the design of a more efficient subspace learning model.

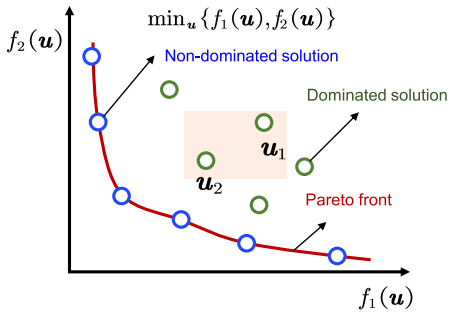


Fig. 1. Illustration of dominated solutions, nondominated solutions, and the Pareto front for a two-objective optimization problem.

B. MultiObjective Optimization

Most real-world optimization problems are essentially MOP, typically formulated as follows:

$$\begin{aligned} &\min\ F(\boldsymbol{u}) = \{f_1(\boldsymbol{u}), f_2(\boldsymbol{u}), \ldots, f_m(\boldsymbol{u})\} \\ &\text{s.t. } t_i(\boldsymbol{u}) = 0, \quad i = 1, \ldots, c_1 \\ &\quad h_i(\boldsymbol{u}) \geq 0, \quad i = 1, \ldots, c_2 \end{aligned} \tag{3}$$

where $f_i()$, $t_i()$, and $h_i()$ denote the objective function, equality constraint, and inequality constraint, respectively, $\boldsymbol{u}$ represents the decision vector, and $\{c_1, c_2\}$ are the number of constraints. Since different objective functions conflict with each other, there is no single optimal solution for MOPs without considering preferences. Instead, there exists a solution set called the Pareto optimal solution set, in which solutions cannot be improved in one objective without deteriorating at least one other objective. Correspondingly, the surface comprising the Pareto optimal set is known as the Pareto front. Furthermore, the concepts of dominated and non-dominated solutions are often used to determine whether one solution is better than another. For example, as depicted in Fig. 1, we say $\boldsymbol{u}_2$ Pareto dominates $\boldsymbol{u}_1$ *iff* $\forall i \in \{1, \ldots, m\} : f_i(\boldsymbol{u}_2) \leq f_i(\boldsymbol{u}_1) \wedge \exists i \in \{1, \ldots, m\} : f_i(\boldsymbol{u}_2) < f_i(\boldsymbol{u}_1)$.

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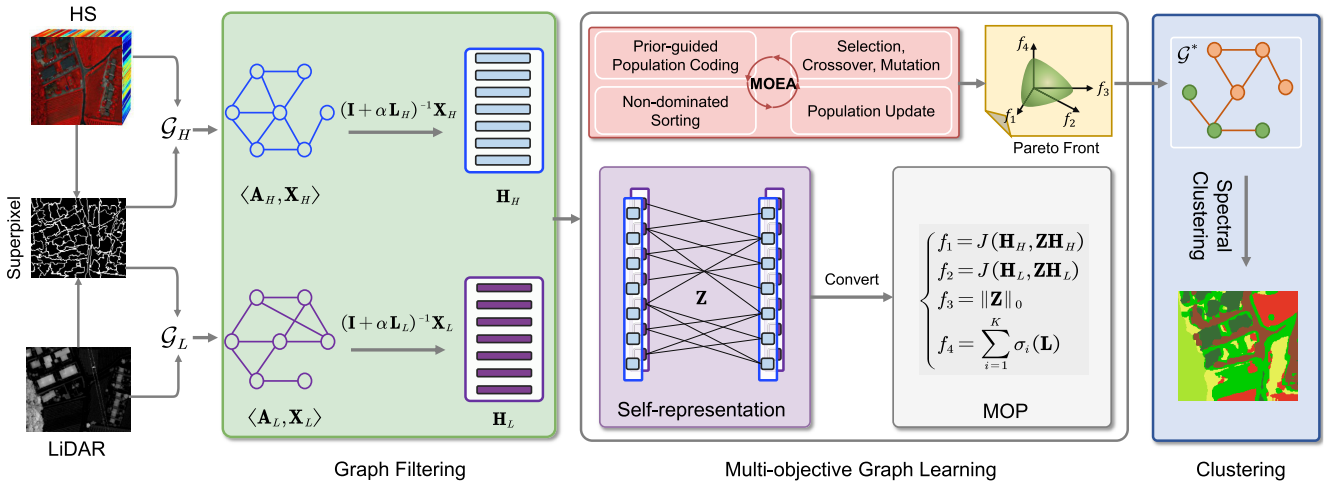


Fig. 2. Framework of MMAGL. Initially, we generate superpixel memberships through the oversegmentation of HS and LiDAR data. Modality-specific attributed graphs $\{\mathcal{G}_H, \mathcal{G}_L\}$ are then constructed separately based on superpixel-level HS and LiDAR features. Subsequently, GF is applied to $\mathcal{G}_H$ and $\mathcal{G}_L$ to smooth features via neighborhood propagation. The acquired representations are jointly utilized for learning a fused graph, defined by a multiobjective system that incorporates four clustering subproblems. An MOEA is employed to solve the system, resulting in the fused optimal graph $\mathcal{G}^*$. Finally, spectral clustering is performed on $\mathcal{G}^*$ to obtain the clustering map.

The MOEA [18] stands out as one of the most effective methods for solving MOP. It can be classified into four categories [33]: decomposition-based, dominance-based, indicator-based, and coevolutionary methods. The decomposition-based method breaks down the target MOP into multiple subproblems, and the solution of each subproblem forms the final solution set, exemplified by MOEA/D [34]. Dominance-based MOEAs adopt a nondominated sorting mechanism along with a diversity maintenance scheme for candidate selection. For instance, NSGA-III [35] uses a crowding distance as a diversity measure in selection after ranking candidate solutions using the fast nondominated sorting method. In indicator-based MOEAs such as HypE [36], candidate solutions that contribute more to the performance indicator are selected. Coevolutionary MOEAs, like TriP [37], implement the concept of cooperation and competition between multiple populations.

Given that many machine learning problems belong to MOPs, there is a growing interest in developing evolutionary machine learning methods [38]. For instance, hybrid subspace clustering methods [19], [20] have been extensively explored in recent years. However, executing MOEAs can be time-consuming, particularly when dealing with large-scale and high-dimensional datasets. Existing hybrid subspace clustering methods directly manipulate a decision vector with a size of N^2. Such an extensive search space limits their applicability to multimodal RS data. To address this challenge, we introduce two efficient strategies by leveraging priors from RS datasets and subspace structures, enabling MOEAs to solve large-scale multimodal RS data clustering problem.

C. Motivation

MVSC fundamentally poses an MOP. Employing MOEAs to address such problems is not only straightforward but also offers several unique advantages, including flexible problem definition, fewer hyperparameters, and the ability to find global solutions. However, it encounters a significant challenge when applied to multimodal RS data, namely, an enormous search space, rendering it impractical.

On one hand, the standard subspace clustering method, modeled by self-representation, requires solving N^2 representation coefficients, leading to combinatorial explosion and substantial storage burden during the search. Although several promising strategies, such as the anchor strategy [13], [13], [21], [39], [40], can significantly reduce complexity, the reduced search space remains substantial for an MOEA method. For instance, with T anchors ($T \leq N$), there are at least $T \times N$ trainable parameters needing to be updated, presenting challenges for large-scale RS scenes.

On the other hand, there are many redundant elements in a self-representation coefficient matrix. First, the ultimate aim of self-representation is to find a suitable similarity matrix, which is usually structured and comprises symmetric elements. Therefore, updating only the upper diagonal elements suffices. Second, the self-representation coefficient matrix is often sparse, with nonzero elements roughly predefined by neighborhood relationships. This implies that a self-representation process can be approximated by adjusting these nonzero elements. Additionally, RS data is often characterized by rich spatial information, presenting as a series of homogeneous regions. This characteristic allows recasting the pixel-level clustering problem into a superpixel-level problem, dramatically reducing the scale of the original problem.

In this article, we introduce a hybrid framework to illustrate the efficient optimization of the MVSC problem using MOEAs. Building on the aforementioned observations, we reformulate MVSC as a multiobjective graph learning problem, aiming to optimize sparse edge weights. Leveraging prior knowledge, we incorporate several strategies to reduce the search space, making it computationally feasible within the context of an MOEA.

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III. Methodology

In this section, we expound upon the proposed method, MMAGL, designed for the joint clustering of HS and LiDAR data. The fundamental concept of MMAGL is depicted in Fig. 2.

A. Superpixel-Level Attributed Graph Generation

Constructing pixel-level graphs for multimodal RS data encounters two challenges. First, pixel-level graphs overlook the local interdependencies of homogeneous regions, which provide crucial spatial information. Second, pixel-level graphs impose a substantial computational burden, especially when dealing with large-scale datasets. In particular, utilizing swarm intelligence optimization algorithms to solve a pixel-level graph becomes computationally prohibitive. To address these issues, we opt for constructing superpixel-level graphs.

As proposed in [25], we employ the entropy rate superpixel segmentation (ERS) to generate homogeneous regions. To fully exploit complementary information, we concatenate HS and LiDAR data along their channel dimension. Subsequently, we apply the ERS method to obtain superpixel memberships. Specifically, denoting the M non-overlapping homogeneous regions as $\mathcal{S} = \{\mathcal{S}_i \big| |\mathcal{S}_i| = n_i,\ i = 1, 2, \ldots, M\}$ where $M \ll N$, and each $\mathcal{S}_i$ contains n_i pixels. We treat $\mathcal{S}$ as the node set of a superpixel-level graph and denote the node attributes as the mean of all pixels in each $\mathcal{S}_i$. Let $\mathbf{Q} \in \mathbb{R}^{M\times N}$ be an association matrix, where $q_{ij} = \mathbb{1}(p_j \in \mathcal{S}_i)$, indicating whether pixel p_j belongs to the superpixel $\mathcal{S}_i$. The node attribute matrix can be calculated by $\mathbf{X} = \mathbf{Q}\mathbf{X}_p$, where $\mathbf{X}_p$ indicates the pixel-level feature matrix.

Furthermore, the edge set $\mathcal{E}$ is calculated using the k-nearest neighbors (kNNs) algorithm, i.e., $\mathcal{E} = \{(\boldsymbol{x}_i, \boldsymbol{x}_j) \big| \boldsymbol{x}_j \in \mathcal{N}_k(\boldsymbol{x}_i) \text{ or } \boldsymbol{x}_i \in \mathcal{N}_k(\boldsymbol{x}_j)\}$. Here, $\mathcal{N}_k(\boldsymbol{x}_i)$ denotes the neighbor set associated with the ith node. Similar to most graph neural networks, we include self-loops in $\mathcal{E}$ to improve the graph representation learning. Finally, based on $\mathcal{S}_H$ and $\mathcal{S}_L$, we obtain two initially attributed graphs, namely, $\mathcal{G}_H = (\mathcal{E}_H, \mathbf{A}_H, \mathbf{X}_H)$ and $\mathcal{G}_L = (\mathcal{E}_L, \mathbf{A}_L, \mathbf{X}_L)$, for HS and LiDAR data, respectively.

It is noteworthy that the computational complexity of the pixel-level graph is reduced from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$ by employing a superpixel-level graph, making it scalable to large hyperspectral images (HSIs). Additionally, the use of homogeneous regions ensures the semantic consistency of clustering results.

B. Graph Filtering

Based on the previously defined attributed graphs, we introduce an efficient GF method to obtain discriminative graph embeddings. Inspired by spectral graph convolution [27], [41], we exemplify this process using $\mathcal{G}_H$. We define the graph signal denoising problem as follows:

$$\arg\max_{\mathbf{H}_H} \frac{1}{2}\|\mathbf{X}_H - \mathbf{H}_H\|_F^2 + \frac{\alpha}{2}\mathrm{Tr}\left(\mathbf{H}_H^\top \mathbf{L}_H \mathbf{H}_H\right) \tag{4}$$

where $\mathbf{H}_H$ represents the graph embedding of HS data, α is a regularization coefficient, and $\mathbf{L}_H$ is the symmetric normalized Laplacian matrix derived from $\mathbf{A}_H$ with the definition of $\mathbf{L}_H = \mathbf{I} - \mathbf{D}_H^{-1/2}\mathbf{A}_H\mathbf{D}_H^{-1/2}$, where $\mathbf{D}_H$ indicates a diagonal degree matrix. By calculating the partial derivative, it is straightforward to verify that (4) has the following closed-form solution:

$$\mathbf{H}_H = (\mathbf{I} + \alpha\mathbf{L}_H)^{-1}\mathbf{X}_H. \tag{5}$$

Let $\mathbf{P}_H = (\mathbf{I} + \alpha\mathbf{L}_H)^{-1}$ be a propagation matrix. Since we have reduced the attributed graph to a suitable scale using superpixels, (5) can be efficiently calculated. Similarly, we perform GF on $\mathcal{G}_L$, yielding $\mathbf{H}_L$ for the LiDAR data.

We attempt to provide a theoretical analysis for the proposed graph filter. Based on Proposition 1, we demonstrate that our graph filter is a more general form of spectral graph convolution, implying better generalization ability. Moreover, we show that our graph filter acts as a low-pass filter. This process enhances the discriminative power of the resulting embedding, thereby facilitating node clustering.

Proposition 1: The vanilla GCN [42] is the first-order approximation of the Taylor expansion for $\mathbf{P}$. Equation (5) acts as a low-pass filter whose frequency response function is $g(\lambda) = 1/(1 + \alpha\lambda)$, where λ is the eigenvalue of a Laplacian matrix $\mathbf{L}$.

Proof: The vanilla GCN without the nonlinear activation is modeled by $\mathbf{H} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{X}$. Its propagation matrix can be rewritten as $\mathbf{P}_{\text{gcn}} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} = \mathbf{I} - \mathbf{L}$. Recall the Taylor series, we have the following relationship:

$$\mathbf{P} = (\mathbf{I} + \alpha\mathbf{L})^{-1} = \sum_{n=0}^{\infty}(-\alpha)^n\mathbf{L}^n \approx \mathbf{I} - \alpha\mathbf{L}. \tag{6}$$

Obviously, the vanilla GCN is a special case of our propagation rule with $\alpha = n = 1$. Similarly, one can easily prove that popular spectral convolutions, e.g., PPNP [43] and SGC [44], belong to the special cases of (5) with different coefficients.

We then prove the low-passing filter. Notice that the Laplacian matrix $\mathbf{L}$ is positive semidefinite, and its eigenvalues λ fall into the interval $[0, 2]$. By treating λ as the frequency of a graph signal, we can define the frequency response function of our filter $(\mathbf{I} + \alpha\mathbf{L})^{-1}$ as

$$g(\lambda) = \frac{1}{1 + \alpha\lambda}, \quad \text{s.t. } \lambda \in [0, 2]. \tag{7}$$

It is easy to check that $g(\lambda)$ is a monotonically decreasing function, and its frequency response is shrunk within $[0, 1]$ when α is fixed. This signifies that only low-frequency signals can pass the filter processing. Furthermore, the regularization coefficient α controls the shrinking rate, with higher α filtering out more high-frequency components. ■

C. Neighborhood Contrastive Self-Representation

Traditional self-representation evaluates reconstruction quality by computing pairwise errors, disregarding interdependencies among data points, which results in heightened sensitivity to outliers. To address this limitation, we introduce a novel NCSR that incorporates positive and negative pairs. Let $\tilde{\mathbf{X}}$ represent the reconstruction corresponding to $\mathbf{X}$, treating them as two distinct views of the data in line with contrastive learning principles [25]. For simplicity, denote $\hat{\mathbf{X}} = [\mathbf{X}, \tilde{\mathbf{X}}]$

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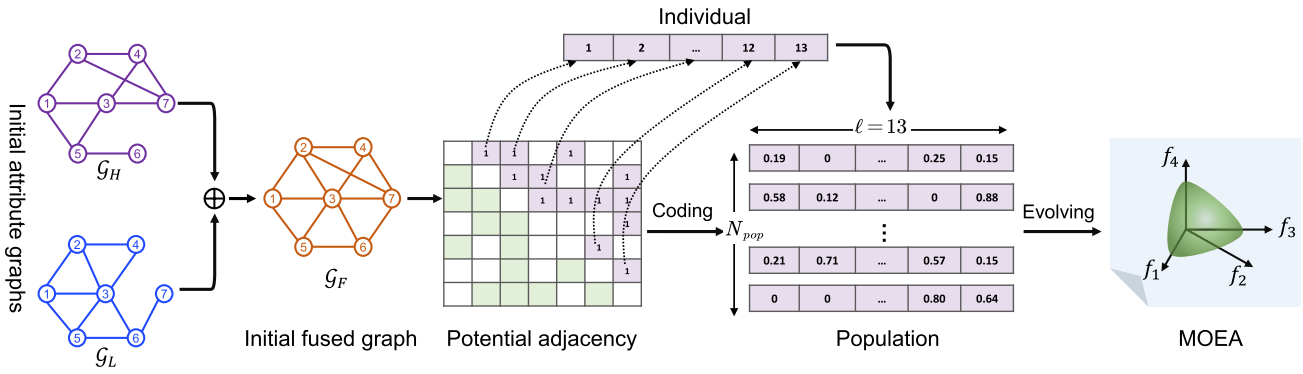


Fig. 3. Schematic representation of the proposed prior-guided population coding strategy.

as the 2 M augmented samples. For each anchor $\tilde{\boldsymbol{x}}_i$, positive pairs associated with it are defined as neighbors in both inputs and reconstructions, i.e., $\{\tilde{\boldsymbol{x}}_i, \hat{\boldsymbol{x}}_j\} : \forall j \in \mathcal{N}(i) \cup \mathcal{N}(\tilde{i})$, and the remaining pairs are considered negative. The optimization problem of NCSR is expressed as follows:

$$J\left(\mathbf{X}, \tilde{\mathbf{X}}\right) = \sum_{i=1}^{M} \sum_{j \in \mathcal{N}(i) \cup \mathcal{N}(\tilde{i})} -\log\left(\frac{\exp\left(\rho\left(\tilde{\boldsymbol{x}}_i, \hat{\boldsymbol{x}}_j\right)/\tau\right)}{\sum_{j \neq i}^{2M} \exp\left(\rho\left(\tilde{\boldsymbol{x}}_i, \hat{\boldsymbol{x}}_j\right)/\tau\right)}\right) \tag{8}$$

where ρ denotes a similarity measure function (e.g., cosine similarity or Euclidean similarity), and τ is a temperature coefficient. In contrast to traditional self-representation, NCSR encourages higher similarity between positive pairs while inhibiting negative pairs. By incorporating interactions between data points, NCSR enhances the discriminative power and robustness of self-representation.

D. Definition of Learning Objectives

Modeling an MVSC problem as an MOP problem is an intuitive approach. One of the significant advantages of this modeling choice is the elimination of manual hyperparameters, which often depend heavily on expert knowledge, including view-specific weights and various regularization coefficients. The objective is to derive a consensus subspace matrix that possesses the following characteristics: minimal cross-modal reconstruction error, favorable sparsity, and block diagonalization. For the joint clustering of HS and LiDAR data, we formulate our MOP as follows:

$$\begin{aligned} &\min_{\mathbf{Z}} F(\mathbf{Z}) = \min_{\mathbf{Z}} \{f_1, f_2, f_3, f_4\} \\ &\text{s.t.} \begin{cases} f_1 = J(\mathbf{H}_H, \mathbf{Z}\mathbf{H}_H) \\ f_2 = J(\mathbf{H}_L, \mathbf{Z}\mathbf{H}_L) \\ f_3 = \|\mathbf{Z}\|_0 \\ f_4 = \sum_{i=1}^{K} \sigma_i(\mathbf{L}). \end{cases} \end{aligned} \tag{9}$$

Here, f_1 and f_2 represent the reconstruction errors for the HS and LiDAR data, respectively, measuring the quality of view-specific NCSR. Instead of utilizing $\mathbf{X}_H$ and $\mathbf{X}_L$, we compute the reconstruction procedure based on the view-specific graph

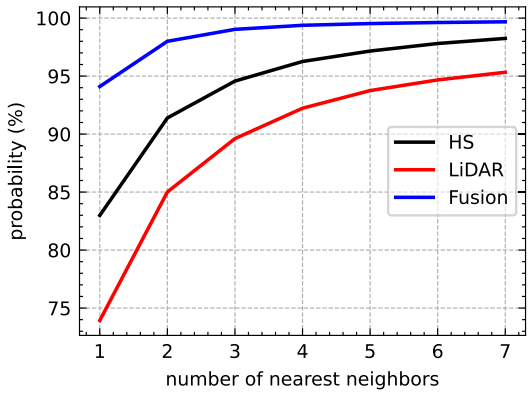


Fig. 4. Probability of involving positive neighbors with varying number of neighbors. We define positive neighbors as those whose labels are the same as the pivot node's label.

embeddings $\mathbf{H}_H$ and $\mathbf{H}_L$ due to their superior discriminative capability. We designate f_3 as a sparsity regularization term, indicating that $\mathbf{Z}$ should contain as few nonzero elements as possible. Notably, we employ a more intuitive L_0 norm, as opposed to the relaxed L_1 or L_2 norms, since MOEAs are adept at handling discrete optimization problems.

Furthermore, following Lemma 1, we introduce f_4 as a Laplacian rank constraint based on the Laplacian matrix $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{Z}\mathbf{D}^{-1/2}$, where $\mathbf{D}$ is computed by $D_{ii} = \sum_j Z_{ij}$. The rationale behind f_4 is that the ideal subspace representation matrix should be block diagonal, indicating that the derived similarity graph has K connected components. As proposed in [23], we convert this hard constraint into a soft version by minimizing the sum of the first K smallest eigenvalues, $\sigma_i(\mathbf{L})$.

Lemma 1: The multiplicity of the eigenvalue zero of the normalized Laplacian matrix $\mathbf{L}$ equals to the number of connected components (and subspaces) in the graph associated with $\mathbf{Z}$.

E. Population Coding and Optimization

Based on the MOP defined in (9), any MOEA can be utilized as the optimizer. MOEAs typically employ mechanisms such as selection, crossover, and mutation to efficiently explore and exploit the search space while maintaining a diverse set of solutions that approximate the Pareto front. Key strategies often include dominance-based

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sorting, diversity preservation techniques, and adaptive parameter control to achieve a balance between convergence and diversity [18], [33].

In this article, we opt for the nondominated sorting genetic algorithm III (NSGA-III) [35] due to its exceptional global searching ability. Unfortunately, as discussed in the motivation, directly optimizing $\mathbf{Z}$ encounters an expansive search space, leading to convergence failures with high computational and storage burdens. This elucidates why fewer MOEA approaches have been applied to real-world large-scale machine learning problems, despite their significant theoretical potential.

Here, we introduce an efficient population coding strategy to circumvent this challenge, referred to as prior-guided coding. The core idea of prior-guided coding is described in Fig. 3. Initially, we define a fused graph by combining the view-specific graphs, denoted as $\mathcal{G}_F = \mathcal{G}_H \oplus \mathcal{G}_L$. We further assume that the potential subspace representation coefficients exist on the pairwise edges of $\mathcal{G}_F$. As shown in Fig. 4, such an assumption is reasonable since the neighbor set of a node usually exhibits an extremely high probability of involving positive pairs. Consequently, our goal is to identify the positive edges in $\mathcal{G}_F$ and assign suitable weights. Let $\mathbf{A}_F$ be the adjacency matrix of $\mathcal{G}_F$, and we treat it as the initial subspace representation. Considering the symmetry of $\mathbf{A}_F$, we only need to focus on the upper diagonal nonzero elements during optimization. Based on these insights, we encode an individual of MOEA as a vector of real-valued variables, $\boldsymbol{u} = \{u_i \in [-1, 1] | i = 1, \ldots, \ell\}$, where ℓ denotes the number of edges of $\mathcal{G}_F$. Let $\mathbf{Z} = \phi(\boldsymbol{u})$ be a function mapping an individual to a subspace representation matrix. Our MOP problem defined in (9) can be rewritten as follows:

$$F(\boldsymbol{u}) = \begin{cases} f_1 = J(\mathbf{H}_H, \phi(\boldsymbol{u})\mathbf{H}_H) \\ f_2 = J(\mathbf{H}_L, \phi(\boldsymbol{u})\mathbf{H}_L) \\ f_3 = \|\boldsymbol{u}\|_0 \\ f_4 = \sum_{i=1}^{K} \sigma_i(\mathbf{L}). \end{cases} \tag{10}$$

Finally, we initialize a group of populations randomly with a size of N_{pop} and globally evolve them through an MOEA method. The resulting Pareto optimal solution set $\mathcal{P}$ is then used to predict the final labels. The pseudocode of MMAGL is shown in Algorithm 1.

Given $\boldsymbol{u}_i^* \in \mathcal{P}$, we construct the optimal fused graph as $\mathcal{G}_i^*$ with corresponding adjacency $\mathbf{A}_i^* = (1/2)(|\phi(\boldsymbol{u}_i^*)| + |\phi(\boldsymbol{u}_i^*)|^\top)$. By applying spectral clustering on $\mathcal{G}_i^*$, we can obtain the superpixel-level labels, denoted as $\boldsymbol{y}_i$. They can be mapped back pixel-level labels by $Y_i = \mathbf{Q}^\top \boldsymbol{y}_i$.

Algorithm 1 Pseudocode of MMAGL

Input: Hyperspectral data $\mathbf{X}^1$, LiDAR data $\mathbf{X}^2$, number of land cover K, hyper-parameters M, α, τ, and hyper-parameters in MOEA.
Output: K clusters

```
Obtain superpixel segmentation S by ERS;
Initialize the attributed graphs G_H and G_L by kNN ;
Conduct graph filtering by Eq. (4);
Encode the initial fused graph as population;
Evolve the fused graph by solve Eq. (10) with MOEA;
Apply spectral clustering on fused optimal graph G*;
```

F. Analysis and Discussion

The proposed method uses superpixels to construct smaller superpixel-level graphs and performs optimization for potential edges, significantly reducing both space and time complexity. To simplify the analysis, we ignore the complexity of preprocessing and superpixel segmentation steps, as they can be completed offline. The main space overhead of MMAGL is to store and process superpixel-level graphs and map superpixel labels to pixels. Let $D = \sum_v D_v$, MMAGL requires at least $\mathcal{O}(M^2 + MD)$ space when computing GF [(5)] and evaluating the objective function [(10)]. For a certain MOEA, its space complexity is mainly determined by the population size. Notably, we only consider nonzero edges, so MMAGL approximately requires $\mathcal{O}(\ell N_{\text{pop}})$ for storing the population, which is far less than processing full graphs or pixel-level graphs. Additionally, superpixel-to-pixel label mapping takes $\mathcal{O}(NM)$ space.

Furthermore, the time complexity of GF is $\mathcal{O}(M^3)$ due to matrix inversion. However, since $M \ll N$, the process is highly efficient. For each generation of MOEA, a single evaluation of the objective takes $\mathcal{O}(M^3 + 8M^2k + \ell)$. For an RS image clustering task, the relationship $k \ll M < \ell$ holds. Thus, the time complexity is dominated by eigen decomposition (i.e., f_4), which takes $\mathcal{O}(M^3)$. Taking NSGA-III as an example, the time complexity for a full evolution iteration is approximately $\mathcal{O}(N_{\text{pop}}M^3 + N_{\text{pop}}^2 + N_{\text{pop}}\ell)$. In practice, one can adjust M to scale MMAGL to large-scale data. Additionally, MOEAs can be easily accelerated by parallel processing, making MMAGL more computationally efficient.

IV. Experiments

In this section, we extensively evaluate our proposed MMAGL on three benchmarking HS-LiDAR datasets and compare its performance against several state-of-the-art approaches. To provide a comprehensive assessment, we analyze five widely used evaluation metrics for each approach.

A. Experiment Setup

1) Datasets:

1) *Trento:* The Trento dataset was captured over a rural area south of the city of Trento, Italy. It comprises HS and LiDAR data with dimensions of 600 ×166 pixels and a spatial sampling of 1 m. The HS data, acquired by the AISA Eagle sensor, consists of 63 spectral bands ranging from 0.40 to 0.98 μm. The LiDAR data, captured by the Optech ALTM 3100EA sensor, contains two channels. The dataset includes six different land-cover types used in the experiment.
2) *MUUFL Gulfport:* The MUUFL Gulfport scene was collected on November 8, 2010, over the University of Southern Mississippi Gulfport Campus, Long

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TABLE I
LAND-COVER TYPES OF THE TRENTO, MUUFL GULFPORT, AND HOUSTON 2013 DATASET

Trento		MUUFL Gulfport		Houston 2013	
Class Name	#Samples	Class Name	#Samples	Class Name	#Samples
Apple trees	4034	Trees	23246	Healthy grass	1251
Buildings	2903	Mostly grass	4270	Stressed grass	1254
Ground	479	Mixed ground surface	6882	Synthetic grass	697
Wood	9123	Dirt and sand	1826	Trees	1244
Vineyard	10501	Road	6687	Soil	1242
Rolds	3174	Water	466	Water	325
		Building shadow	2233	Residential	1268
		Building	6240	Commercial	1244
		Sidewalk	1385	Road	1252
		Yellow curb	183	Highway	1227
		Cloth Panels	269	Railway	1235
				Parking Lot 1	1233
				Parking Lot 2	469
				Tennis Court	428
				Running Track	660
Total	30214	Total	53687	Total	15029

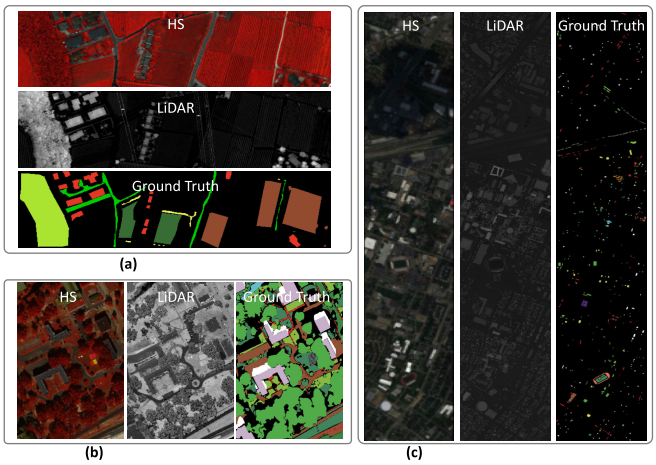


Fig. 5. False-color visualizations of HS modality, LiDAR modality, and ground truth for three datasets. Black in ground-truth denotes unknown classes for each ground truth image. (a) Trento dataset. (b) MUUFL Gulfport dataset. (c) Houston 2013 dataset.

Beach, Mississippi, USA. The HS data was acquired by the ITRES Research Ltd. (ITRES) Compact Airborne Spectrographic Imager (CASI)-1500 sensor. It has dimensions of 325 × 220 pixels with 72 spectral channels (64 available bands) ranging from 375 to 1050 nm at a spectral sampling of 10 nm. The spatial resolution of the hyperspectral data is 0.54 × 1.0 m, while that of the LiDAR-based DSM is 0.60 × 0.78 m. The LiDAR data was acquired by the Gemini Airborne Laser Terrain Mapper (ALTM) LiDAR sensor. There are 11 land-cover categories investigated in the scene.

3) *Houston:* The Houston dataset was acquired over the area of the University of Houston campus and neighboring areas in June 2012. It was released for the IEEE GRSS DFC 2013. The dataset consists of HS data and LiDAR-derived digital surface model (DSM) data, both with dimensions of 349 ×1905 pixels and a spatial resolution of 2.5 m. The HS modality contains 144 spectral bands covering the spectral range from 0.38 to 1.05 μm, and the LiDAR modality contains one channel. There are 15 distinguishable land cover categories investigated in the scene.

The details of the land-cover types for these datasets are listed in Table I, and their visualizations are shown in Fig. 5.

2) Baselines: We conduct experiments on multiple benchmark datasets to compare the following state-of-the-art MVC algorithms.

1) *MVKM [9]:* This method is a classical MVC approach, an extension of k-means based on the co-EM framework.
2) *LMVSC [21]:* A recent representative fast MVC algorithm that learns anchor graphs instead of fully connected graphs of samples for MVSC. It efficiently clusters multiview data with linear computational complexity.
3) *FPMVS-CAG [13]:* Similar to EOMSC-CA, this method jointly conducts anchor selection and subspace graph construction in a unified optimization formulation. It has linear time complexity with respect to the sample number.
4) *SMVSC [40]:* Another representative approach based on the unified anchor graph.
5) *BMVC [45]:* A popular clustering framework for large-scale multimodal data. It effectively integrates compact collaborative discrete representation learning and binary clustering structure learning, making it computationally efficient.
6) *SFMC [23]:* A state-of-the-art graph fusion framework that seeks a joint graph compatible across multiple views in a self-supervised weighting manner. It achieves stable performance and scales well with the data size.
7) *MSGL [46]:* This method introduces a scalable bipartite graph-based MVSC framework that can be generalized to unseen data points.
8) *MCGC [17]:* A novel multiview graph clustering method that incorporates GF and contrastive loss. We customize it for our datasets by leveraging superpixel graphs.
9) *AMKSC [16]:* A unified framework for multimodal RS clustering based on multiview anchor subspace clustering with spatial regularization and linear kernel.

3) Implementation Details: We implemented the proposed MMAGL using MATLAB and conducted all the experiments on an Intel Core i9 13900K with 64 GB memory. As the optimizer, we employed NSGA-III [35] implemented in PlatEMO [47] due to its efficacy in solving MOPs. The population size and the maximum number of function evaluations were set to 50 and 10 000, respectively. For oversegmentation, we used 300, 600, and 700 superpixels for the Trento, MUUFL, and Houston datasets, respectively. The initialization parameters for obtaining graph embedding are set to $k = 7$ and $\alpha = 10$ for all experiments.

4) Metrics: We employ five metrics to quantitatively assess the joint clustering results: overall accuracy (ACC), Kappa coefficient (Kappa), normalized mutual information (NMI), adjusted rand index (ARI), and clustering purity (Purity). Higher values for these metrics indicate better clustering performance. Additional details about these metrics can be found in [14] and [24].

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TABLE II
COMPARISON WITH STATE-OF-THE-ART MULTIMODAL CLUSTERING METHODS ON THE TRENTO DATASET

Class	MVKM	LMVSC	FPMVS-CAG	SMVSC	BMVC	SFMC	MSGL	MCGC	AMKSC	MMAGL
Apple trees	0.7856	0.7395	0.2380	0.3109	0.7397	0.9933	0.4849	1.0000	0.9660	0.7600
Buildings	0.8815	0.9153	0.0000	0.0000	0.0000	0.7981	0.6962	0.6535	0.7261	0.8450
Ground	0.0021	0.8894	0.0772	0.1336	0.0605	0.0000	0.0000	0.0689	0.0042	0.0000
Wood	0.8565	0.9724	0.9838	0.9838	0.9773	0.9898	0.9927	1.0000	0.9849	0.9905
Vineyard	0.5198	0.5948	0.7305	0.6862	0.4473	0.8426	0.9950	0.8371	0.9711	0.9634
Rolds	0.5803	0.6890	0.9049	0.9061	0.8885	0.7202	0.2990	0.7987	0.4200	0.7776
ACC	0.6898	0.7735	0.6790	0.6744	0.6436	0.8767	0.8086	0.8742	0.8778	**0.8983**
Kappa	0.6157	0.7138	0.5715	0.5746	0.5499	0.8380	0.7393	0.8345	0.8361	**0.8635**
NMI	0.6256	0.6778	0.5320	0.5651	0.6060	0.7779	0.7740	0.7871	0.7824	**0.8152**
ARI	0.5330	0.6488	0.5334	0.5780	0.5524	0.8063	0.7914	0.8108	0.8448	**0.8521**
Purity	0.7683	0.8176	0.7202	0.7385	0.7547	0.8994	0.8119	0.8866	0.9113	**0.9094**
CPU Time	2.777	124.972	1304.906	787.634	156.035	342.603	1374.061	15.554	37.523	62.193

TABLE III
COMPARISON WITH STATE-OF-THE-ART MULTIMODAL CLUSTERING METHODS ON THE MUUFL DATASET

Class	MVKM	LMVSC	FPMVS-CAG	SMVSC	BMVC	SFMC	MSGL	MCGC	AMKSC	MMAGL
Trees	0.3797	0.1692	0.4757	0.4724	0.5051	0.5118	0.5225	0.6442	0.4544	0.7095
Mostly grass	0.7288	0.0567	0.0337	0.0487	0.7030	0.1848	0.3073	0.4309	0.7536	0.5820
Mixed ground surface	0.4954	0.5648	0.0702	0.1562	0.4801	0.2043	0.3987	0.2865	0.2326	0.4462
Dirt and sand	0.0022	0.4261	0.0077	0.0142	0.0049	0.0148	0.0000	0.5011	0.1659	0.6972
Road	0.6390	0.5280	0.5823	0.5693	0.7390	0.6623	0.7387	0.4086	0.7670	0.6380
Water	0.0000	0.1717	0.0021	0.0000	0.1288	0.8541	0.0601	0.4335	0.8155	0.0215
Building shadow	0.8159	0.4975	0.8889	0.8943	0.2163	0.1039	0.8988	0.5320	0.3193	0.3072
Building	0.2622	0.3598	0.1244	0.1138	0.4104	0.3450	0.2002	0.5014	0.4997	0.5248
Sidewalk	0.1509	0.1148	0.0159	0.0152	0.0152	0.0022	0.0022	0.0852	0.1307	0.0729
Yellow curb	0.0383	0.0000	0.0055	0.1421	0.0219	0.0000	0.0929	0.0000	0.2678	0.0109
Cloth Panels	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.9926	0.0818	0.0000
ACC	0.4340	0.2974	0.3423	0.3511	0.4867	0.3974	0.4553	0.5092	0.4708	**0.5897**
Kappa	0.3473	0.2289	0.1878	0.2052	0.3745	0.2645	0.3517	0.4032	0.3864	**0.4921**
NMI	0.4108	0.3340	0.2767	0.2844	0.3792	0.3080	0.4414	0.3893	0.4118	**0.4360**
ARI	0.2572	0.1373	0.1955	0.1973	0.2846	0.1564	0.3014	0.3926	0.2902	**0.4713**
Purity	0.6510	0.6187	0.5154	0.5285	0.6195	0.5346	0.6391	0.6189	0.6512	**0.6674**
CPU Time	489.148	33.646	456.243	268.949	27.002	237.040	1210.495	24.722	38.033	989.029

B. Main Results

We conducted comprehensive experiments to compare our proposed MMAGL method with nine state-of-the-art MVC baselines on three real HS-LiDAR datasets. The experimental results, including per-class accuracy and five clustering indices (OA, Kappa, NMI, ARI, and Purity), along with computational time, are presented in Tables II–IV. The best results are highlighted in boldface for better visibility. Additionally, clustering maps generated by different methods are provided in Figs. 6–8 for visual comparison.

Tables II–IV demonstrate that the proposed MMAGL consistently outperforms existing methods across all clustering metrics. Specifically, MMAGL demonstrates substantial improvements over other methods, with notable margins. For example, it exceeds the performance of the state-of-the-art method (AMKSC) by 2.05%, 11.89%, and 5.08% in terms of ACC on the Trento, MUUFL, and Houston datasets, respectively. When analyzing per-class accuracy, MMAGL achieves more uniform results compared to its competitors. Notably, Table IV reveals that FPMVS-CAG and SMVSC exhibit significant clustering failures (zero accuracy) for certain land-cover types, such as Healthy grass, Stressed grass, and Synthetic grass, indicating their inability to capture the intrinsic subspace structure and reveal fine-grained differences in land-cover types.

Furthermore, MMAGL's CPU time is comparable to LMVSC and BMVC and faster than most of the baselines. This efficiency is attributed to the strategies of superpixel-level graphs and prior-based population coding, which significantly reduce the search space. Despite the success of anchor-based strategies employed by all compared MVSC methods, their efficiency can be further improved by adopting our superpixel-based strategy.

From Figs. 6–8, it is evident that, by leveraging the superpixel graph, MMAGL and MCGC achieve more homogeneous clustering results compared to the other methods. In contrast, results obtained by MVKM, LMVSC, EOMSC-CA, FPMVS-CAG, and BMVC exhibit considerable salt-and-pepper noise, leading to poor smoothing. This observation indicates that MMAGL effectively learns correct semantic and spatial structures for different land-cover types.

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TABLE IV
COMPARISON WITH STATE-OF-THE-ART MULTIMODAL CLUSTERING METHODS ON THE HOUSTON 2013 DATASET

Class	MVKM	LMVSC	FPMVS-CAG	SMVSC	BMVC	SFMC	MSGL	MCGC	AMKSC	MMAGL
Healthy grass	0.8577	0.8537	0.0000	0.0000	0.8161	0.2062	0.8513	0.5795	0.8417	0.7370
Stressed grass	0.0000	0.0478	0.0000	0.0000	0.1284	0.0805	0.1994	0.2129	0.2169	0.2033
Synthetic grass	0.7260	0.7604	0.0000	0.0000	0.6298	1.0000	0.8895	0.8034	0.4864	0.7948
Trees	0.6117	0.2082	0.8979	0.8891	0.9244	0.4968	0.1061	0.3834	0.8441	0.5836
Soil	0.9767	0.9662	0.8671	0.8325	0.9919	0.4219	0.7673	0.9766	0.3913	0.7552
Water	0.0000	0.0000	0.0000	0.0738	0.1046	0.7077	0.0000	0.0000	0.2954	0.0000
Residential	0.4866	0.0670	0.0197	0.0418	0.1262	0.2926	0.4913	0.5032	0.0032	0.3470
Commercial	0.2331	0.3272	0.0088	0.0394	0.2275	0.1809	0.1166	0.0233	0.1117	0.1809
Road	0.0040	0.4377	0.0471	0.1893	0.1701	0.1006	0.0958	0.0519	0.3738	0.1709
Highway	0.0000	0.3464	0.0057	0.0057	0.0000	0.0676	0.2657	0.5762	0.3562	0.2070
Railway	0.5538	0.4138	0.8988	0.7919	0.5838	0.6785	0.6656	0.3198	0.6138	0.7660
Parking Lot 1	0.0000	0.4079	0.0373	0.0373	0.4161	0.0543	0.4031	0.0852	0.2368	0.2522
Parking Lot 2	0.0000	0.0064	0.0000	0.0235	0.0597	0.2047	0.0000	0.1002	0.0000	0.1898
Tennis Court	0.6402	0.6215	0.7313	0.8084	0.4089	0.6682	0.0000	0.9673	0.0000	0.9907
Running Track	0.0000	0.0030	0.8000	0.8424	0.0394	0.0000	0.3227	0.5636	0.6970	0.4833
ACC	0.3608	0.3904	0.2856	0.2959	0.4096	0.3008	0.3837	0.4002	0.3895	**0.4403**
Kappa	0.3084	0.3424	0.2246	0.2372	0.3627	0.2527	0.3322	0.3557	0.3429	**0.3972**
NMI	0.4140	0.4519	0.3989	0.4148	0.4654	0.3581	0.4655	0.4347	0.4283	**0.4803**
ARI	0.2080	**0.2826**	0.1764	0.1853	0.2689	0.1730	0.2583	0.2453	0.2501	0.2721
Purity	0.3688	0.4187	0.3028	0.3222	0.4467	0.3300	0.3985	0.4193	0.4144	**0.4738**
CPU Time	1172.478	651.789	10121.024	8829.306	830.908	4044.199	10494.098	82.157	105.355	632.746

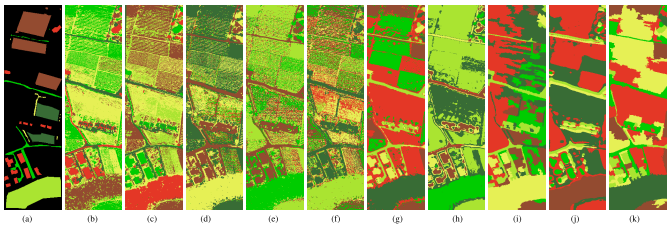


Fig. 6. Clustering maps of different multimodal clustering approaches on the Trento dataset. (a) GT. (b) MVKM. (c) LMVSC. (d) FPMVS-CAG. (e) SMVSC. (f) BMVC. (g) SFMC. (h) MSGL. (i) MCGC. (j) AMKSC. (k) MMAGL.

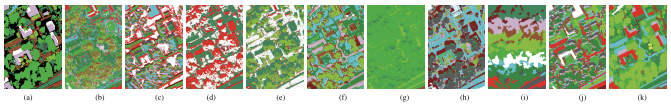


Fig. 7. Clustering maps of different multimodal clustering approaches on the MUUFL Gulfport dataset. (a) GT. (b) MVKM. (c) LMVSC. (d) FPMVS-CAG. (e) SMVSC. (f) BMVC. (g) SFMC. (h) MSGL. (i) MCGC. (j) AMKSC. (k) MMAGL.

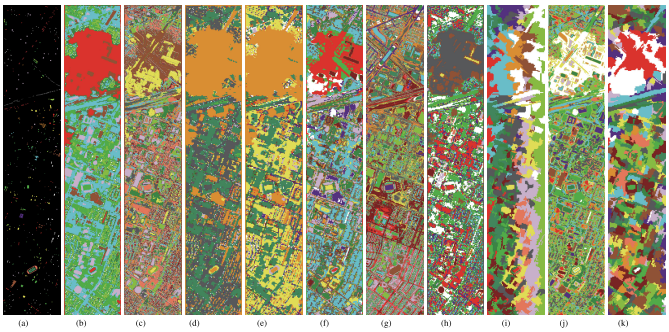


Fig. 8. Clustering maps of different multimodal clustering approaches on the Houston dataset. (a) CCA. (b) MVKM. (c) LMVSC. (d) FPMVS-CAG. (e) SMVSC. (f) BMVC. (g) SFMC. (h) MSGL. (i) MCGC. (j) AMKSC. (k) MMAGL.

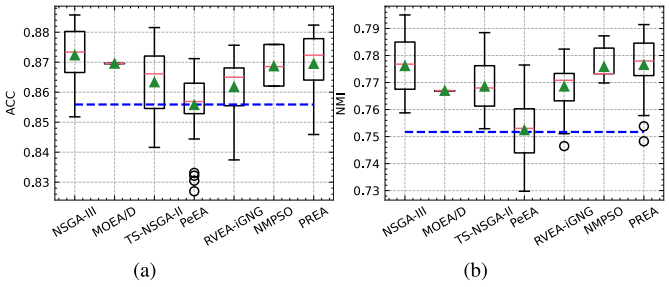


Fig. 9. Performance of different MOEAs on the Trento dataset. We plot the box of the results obtained by the final Pareto solution set for each method. The blue dotted line denotes the baseline performance without optimization. (a) ACC. (b) NMI.

C. Ablation Studies and Parameter Sensitivity

In this section, we thoroughly investigate the behaviors of the introduced components of the proposed MMAGL and assess the performance under various parameter settings.

1) Influence of Different MOEAs: To assess the universality of the proposed method, we replaced the optimizer in MMAGL with seven widely used MOEAs: NSGA-III [35], MOEA/D [34], TS-NSGA-II [48], PeEA [49], RVEA-iGNG [50], NMPSO [51], and PREA [52]. Box plots representing the performance of different optimizers are shown in Fig. 9(a) and (b), where each box corresponds to the clustering results associated with all nondominated solutions obtained by the optimizer. In these plots, the blue line represents the initial result, the green points indicate the averaged performance, and the red lines represent the median.

Observing Fig. 9(a) and (b), we note that the averaged performance and the median of all optimizers are significantly higher than the initial result. Among the different optimizers,

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TABLE V

PERFORMANCE WITH DIFFERENT GRAPHS. VALUES IN () DENOTE THE NUMBER OF EDGES OF CORRESPONDING GRAPHS

Dataset	Graph	ACC	Kappa	NMI	ARI	Purity
Trento	$\mathcal{G}_H(2770)$	0.8310	0.7811	0.7830	0.8134	0.8845
	$\mathcal{G}_L(2740)$	0.6315	0.5258	0.5746	0.5491	0.7430
	$\mathcal{G}_F(2514)$	0.8559	0.8090	0.7517	0.7928	0.8823
	$\mathcal{G}^*(2473)$	**0.8983**	**0.8635**	**0.8152**	**0.8521**	**0.9094**
Houston	$\mathcal{G}_H(2634)$	0.4009	0.3557	0.4709	0.2455	0.4455
	$\mathcal{G}_L(2630)$	0.3264	0.2835	0.3886	0.1736	0.3642
	$\mathcal{G}_F(2537)$	0.4176	0.3755	0.4815	0.2641	0.4602
	$\mathcal{G}^*(2497)$	**0.4403**	**0.3972**	**0.4803**	**0.2721**	**0.4738**

TABLE VI

CLUSTERING RESULTS ON THREE-MODALITY HOUSTON 2013 DATASET (HS + LIDAR + MS)

Method	ACC	Kappa	NMI	ARI	Purity
MVKM	0.3610	0.3086	0.4126	0.2070	0.3685
LMVSC	0.4108	0.3633	0.4765	0.2715	0.4676
BMVC	0.4104	0.3651	0.4498	0.2596	0.4480
MSGL	0.3471	0.2961	0.4565	0.2546	0.3556
AMKSC	0.4091	0.3634	0.4334	0.2321	0.4286
MMAGL(HS+LiDAR)	0.4403	0.3972	0.4803	0.2721	0.4738
MMAGL(HS+LiDAR+MS)	**0.4934**	**0.4546**	**0.5277**	**0.3179**	**0.5197**

NSGA-III and PREA tend to achieve better results on the Trento dataset (e.g., the highest and averaged performance), while MOEA/D and NMPSO exhibit more stable performance compared to others. This experiment underscores the effectiveness of leveraging MOEAs to address the MVSC problem. As a flexible framework, any existing MOEA can be seamlessly integrated into the proposed MMAGL, emphasizing the method's adaptability and generalizability.

2) Ablation Study of Modalities: We conducted experiments on four types of attributed graphs to assess their significance in our method. These graphs include the hyperspectral graph ($\mathcal{G}_H$), LiDAR graph ($\mathcal{G}_L$), initial fused graph ($\mathcal{G}_F$), and optimal fused graph ($\mathcal{G}^*$). The results are presented in Table V. It is evident that fused graphs yield considerable improvements over single-modality graphs, underscoring the complementarity between different modalities. When examining individual modalities, the hyperspectral graph outperforms the LiDAR graph significantly, attributed to the richer discriminative information provided by hyperspectral data.

Comparing the initial fused graph ($\mathcal{G}_F$) with the evolved optimal graph ($\mathcal{G}^*$) obtained by the MOEA, we observe substantial improvements achieved by $\mathcal{G}^*$. For example, $\mathcal{G}^*$ exhibits improvements over $\mathcal{G}_F$ by margins of 4.34% and 2.27% in terms of ACC on the Trento and Houston datasets, respectively. This highlights the efficacy of MMAGL in refining and enhancing the initial fused graph. Despite $\mathcal{G}_F$ having approximately twice the number of edges compared to $\mathcal{G}_H$ or $\mathcal{G}_L$ due to partly overlapped neighbors, MMAGL successfully removes redundant edges in $\mathcal{G}^*$, resulting in a more informative graph. After optimization, 41 and 40 edges are removed in $\mathcal{G}^*$ for the Trento and Houston datasets, respectively, leading to more robust and convincing cluster assignments.

3) Generalization on Three-Modality Data: To evaluate the robustness and scalability of the proposed method, we extended MMAGL to the three-modality Houston dataset by incorporating the multispectral (MS) modality. The experimental results are presented in Table VI. We observe that MMAGL with three modalities (HS + LiDAR + MS) achieves a significant improvement (+5.31% ACC) compared to its performance with two modalities (HS + LiDAR) and outperforms other three-modality baselines significantly. This demonstrates that MMAGL is both flexible and scalable with varying modalities. However, as the number of modalities increases, the optimization problem becomes a complex many-objective optimization challenge, necessitating the use of more efficient optimizers.

TABLE VII

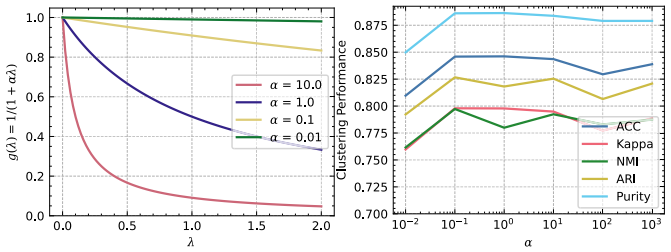
RESULTS OF ABLATION STUDY ON THE OBJECTIVE FUNCTIONS AND GF

Data	Metric	w/o f_1	w/o f_2	w/o f_3	w/o f_4	w/o NCSR	w/o GF	MMAGL
Trento	ACC	0.8863	0.8885	0.8959	0.8904	0.8867	0.8686	**0.8983**
	Kappa	0.8471	0.8500	0.8603	0.8533	0.8475	0.8238	**0.8635**
	NMI	0.7981	0.7910	0.8103	0.7928	0.8034	0.7754	**0.8152**
	ARI	0.8312	0.8372	0.8506	0.8539	0.8345	0.7969	**0.8521**
	Purity	0.8975	0.8996	0.9071	0.9016	0.8971	0.8801	**0.9094**
Houston	ACC	0.4117	0.4133	0.4109	0.4380	0.4224	0.4209	**0.4403**
	Kappa	0.3689	0.3691	0.3680	0.3984	0.3799	0.3792	**0.3972**
	NMI	0.4427	0.4564	0.4574	0.4626	0.4413	0.4710	**0.4803**
	ARI	0.2218	0.2566	0.2556	0.2694	0.2452	0.2462	**0.2721**
	Purity	0.4362	0.4381	0.4467	0.4565	0.4458	0.4503	**0.4738**

4) Ablation Study of Objective Functions and GF: To assess the effectiveness of the proposed objective functions, we conducted experiments by individually removing each sub-objective. Additionally, we explored the impact of NCSR by replacing it with traditional self-representation. The results of the ablation study are presented in Table VII, highlighting the varying influences of the four sub-objectives (f_1, f_2, f_3, and f_4) on the clustering results. Removing f_4 results in the smallest decrease in MMAGL's performance compared to the removal of the other three sub-objectives. This is likely due to the inherent difficulty of optimizing an eigenvalue problem with an MOEA. Since hyperspectral data contributes more than LiDAR data, the removal of f_1 leads to a more significant decrease in performance compared to the removal of f_2. Although f_3 encourages a sparser subspace representation, its impact is found to be less significant than that of NCSR. The proposed NCSR demonstrates a substantial impact on clustering performance for both datasets. It achieves improvements of 1.16%–1.79% over traditional self-representation on the two datasets, respectively, showcasing its superiority. Introducing contrastive learning through NCSR allows traditional self-representation to explore a more complex data distribution, resulting in enhanced clustering performance.

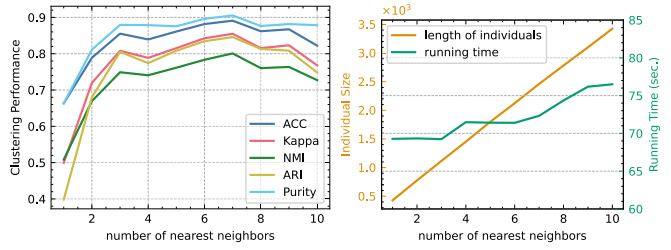
In this experiment, we also conduct an ablation study on the GF to demonstrate its effectiveness. From Table VII, we observe that the ACC drops by 2.97% on the Trento dataset and 1.94% on the Houston dataset when GF is removed. To further investigate the behavior of GF, we present the frequency response function and the relationship between α and λ in Fig. 10(a). Notably, $g(\lambda)$ behaves as a low-frequency filter, restricting the frequency response to the interval [0, 1]. In the context of a graph, low-frequency components of a node are determined by its neighboring nodes. As a result, our

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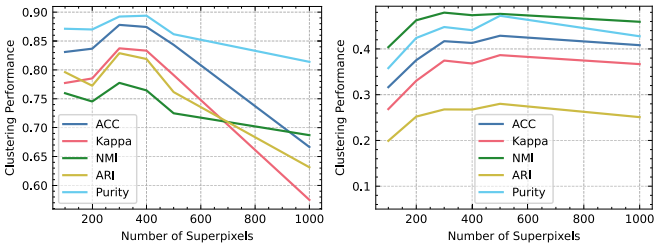
(a) (b)

Fig. 10. (a) Frequency responses of the proposed GF. (b) Clustering performance with varying α on the Trento dataset.



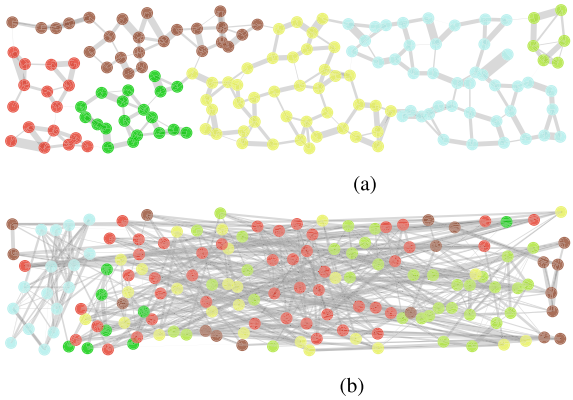
(a) (b)

Fig. 11. Clustering performance and running time with varying number of neighbors in the initial graphs. (a) Clustering performance. (b) Size of individual and running time.



(a) (b)

Fig. 12. Clustering performance with different numbers of superpixels. (a) Trento. (b) Houston.



(a)

(b)

Fig. 13. Visualization of optimized graphs obtained on Trento dataset. Thicker edges denote higher subspace coefficients. (a) Subspace is initialized with centroid-based spatial neighbors. (b) Subspace is initialized with Euclidean distance-based similarity graph.

graph filter essentially performs neighbor aggregation, which is in line with the principles of spectral graph convolution. Furthermore, the parameter α controls the rate of shrinkage, with higher values of α resulting in the filtration of more high-frequency components. In theory, a larger α emphasizes highly similar information, thereby facilitating subsequent subspace learning. The trend observed in Fig. 10(b) shows that larger values of α generally lead to more stable clustering performance, which aligns well with the theoretical expectations of GF.

5) Effect of the Size of Neighbors: In Fig. 11(a), the performance of MMAGL under different numbers of neighbors in the initial graphs is presented. The clustering performance of MMAGL increases with a greater number of nearest neighbors. The empirical study in Fig. 4 suggests that using a larger number of neighbors may enhance the proportion of true positive pairs in the candidate variables, thereby improving robustness. However, larger numbers of neighbors not only introduce more redundant edges in the graphs but also enlarge the search space for MOEAs. In Fig. 11(b), the size of the individual linearly increases with the number of neighbors, accompanied by a slight increase in running time. Excessively large neighbor values can lead to poor convergence during optimization. To strike a balance between performance and efficiency, it is advisable to set $k = 7$ for most datasets in our experiments.

6) Effect of the Number of Superpixels: In our exploration of the sensitivity of the number of superpixels, we varied this parameter within the range of $\{100, 200, \ldots, 1000\}$. As depicted in Fig. 12, the best results for MMAGL were achieved when the number of superpixels was set to 300 for the Trento dataset and 400 for the Houston dataset. However, the performance of MMAGL sharply decreased when the number of superpixels exceeded 400 in Fig. 12(a). Conversely, segmenting the Houston dataset into more superpixels tended to yield steady performance, as shown in Fig. 12(b). This discrepancy can be attributed to differences in resolution and the distribution of land-cover objects. For instance, the Houston dataset comprises more complex land-cover types, necessitating a more detailed segmentation. Generally, finding the globally optimal value for this hyperparameter is challenging. It is important to note that increasing the number of superpixels will exponentially expand the search space, adding to the computational complexity of the optimization process.

7) Visualization of Optimal Graph: While numerous strategies can be employed to initialize the subspace representation of MMAGL, their performance can vary significantly. In this context, we investigate two distinct graph-based initializations on the Trento dataset: centroid-based spatial graphs and Euclidean distance-based similarity graphs. Both graphs consider three neighbors and consist of 150 nodes. In Fig. 13, we visually represent the optimized graphs, with each node denoting a superpixel and different clusters marked in colors. Despite the spatial graph-based strategy achieving better spatial consistency, its clustering ACC is substantially lower (0.6755) compared to that obtained by the similarity graph (0.8602). This observation might appear counterintuitive, but it can be rationalized by considering the characteristics of RS data. After superpixel segmentation, spatially neighboring nodes often belong to different homogeneous regions, possibly even different land-cover types, making it challenging to use these nodes for effective self-representation. On the other

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hand, the similarity graph considers long-range relationships, increasing the likelihood of nodes belonging to the same clusters and proving more beneficial for self-representation. Furthermore, we observe that some edges in the optimized graphs are deactivated, while others are strengthened. Particularly, there are several distinct connected components in Fig. 13(a), a characteristic ensured by the sub-objective f_3. It should be noted that the precision of superpixel segmentation may limit the performance of a superpixel-based method.

V. Conclusion

This article introduces MMAGL, a novel MVC approach designed for HS and LiDAR data. The method employs MOEAs to effectively address the MVC problem. The proposed HS-LiDAR clustering task is formulated as a four-objective attributed graph learning problem, encompassing view-specific self-representations, sparsity, and a rank constraint. To streamline the optimization process, we leverage superpixel segmentation for constructing superpixel-level graphs, exploiting spatial homogeneity to significantly reduce the sample size. Additionally, an efficient population coding strategy is introduced, capitalizing on the subspace distribution's structural prior. These strategies enable MMAGL to be evolutionarily solved through MOEAs. Moreover, a generalized GF operation is presented, theoretically validated as a low-pass filter, along with an NCSR that considers negative pair-wise relationships. Both enhancements contribute to the discriminative capability of MMAGL. Extensive experiments on three real HS-LiDAR datasets demonstrate that our method achieves competitive performance compared to state-of-the-art methods. Particularly, the application of various popular MOEAs consistently yields significant improvements over baselines. The success of MMAGL introduces a new framework for designing MVC methods based on multiobjective optimization.

However, several limitations in MMAGL remain. For instance, MMAGL does not fully exploit the properties of the Pareto solution set, and its performance might impact by superpixel segmentation. In future work, further exploration of Pareto solution selection methods, such as ensemble approaches based on the Pareto solution set, as well as the adoption of more advanced superpixel segmentation techniques, will be considered to enhance the robustness and efficiency of the method in practical applications.

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